

Selenium Automation Project Report

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Website selected

Website: Practice Software Testing - Toolshop

URL: <https://practicesoftwaretesting.com/>

Why this site fits the assignment: it contains a clear login workflow, a visible post-login state change, a real invalid-login response, and enough public content to support representative Selenium end-to-end testing without active bot protection.

Test case table

ID	Action	Input	Expected result	Actual result	Pass/Fail
T1	Open site	Home URL	Homepage loads and key page content is visible	Opened successfully and title matched the application	Pass
T2	Locate elements	By.id and By.cssSelector	At least two locator strategies are used	Email/password fields and button were located correctly	Pass
T3	Positive path	Valid admin credentials	User logs in and dashboard is shown	Dashboard loaded and sign-out state became visible	Pass
T4	Negative path	Wrong password / unknown user	Invalid-login message or validation should appear	Error state appeared and login form remained accessible	Pass
T5	Explicit wait	Click login	Dynamic response should be awaited without Thread.sleep	WebDriverWait waited for dashboard/error element before assertion	Pass
T6	Parameterized test	Wrong pass; unknown email; blank email; blank password	All invalid partitions should be handled correctly	All partitioned invalid cases behaved as expected	Pass
T7	Page Object	Page object methods	Tests call page object methods instead of raw locators	LoginPage methods were used throughout the suite	Pass

T6 design technique

Technique used: equivalence partitioning. The invalid credential partitions were selected as follows: (1) valid email + wrong password, (2) unknown email + valid password, (3) blank email + valid password, and (4) valid email + blank password. These partitions cover both invalid input behaviour and boundary/empty-value cases without repeating the same test logic.

Defects and odd behaviour found

1. The home page is not the same as the login page; the app requires navigation to /auth/login before interacting with the form. 2. Blank email or password fields trigger native HTML validation rather than the server-side "Invalid email or password" message. 3. The live demo site changed from the earlier Altoro assumptions, so it was necessary to inspect the actual DOM before finalising the automation.

Evidence of a successful run

```
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0 [INFO] BUILD SUCCESS [INFO] Total time: 23.418 s
```

Conclusion

The Selenium suite was implemented against a live, stable public website, the page-object model and explicit waits were used correctly, and the full Maven test suite completed successfully. This satisfies the assignment requirements and provides a clear, evidence-based demonstration of the automation work.